GATE 2027

Data Science & Artificial Intelligence (DA)

PREDICTED MOCK PAPER — 2

Above GATE level

Duration: 3 hours **Maximum marks: 100** **65 questions**

General instructions

1. The paper has two sections: **General Aptitude (GA)** — Q.1 to Q.10, 15 marks — and the **core Data Science & AI** section — Q.11 to Q.65, 85 marks.
2. Q.1–Q.5 and Q.11–Q.35 carry **one mark** each. Q.6–Q.10 and Q.36–Q.65 carry **two marks** each.
3. Each question is of one of three types, marked next to the question number:

Type	Description	Negative marking
MCQ	four choices, exactly one is correct	$-1/3$ (1-mark), $-2/3$ (2-mark)
MSQ	four choices, one or more are correct; no partial credit	none
NAT	a numerical value, to be entered directly	none

4. Statistical tables for the standard normal, t and χ^2 distributions, where needed, are the same as those supplied in the accompanying question bank.
5. The answer key appears at the end of this booklet. **Do not consult it until the three hours are over.**

How this paper was constructed. The subject-wise mark split, the MCQ/MSQ/NAT mix and the one-mark/two-mark balance all follow the average of the GATE DA 2024, 2025 and 2026 papers. Paper 2 is deliberately pitched *above* the observed level — more multi-step 2-mark questions and more MSQs — as a hedge against a harder 2027 paper. See the accompanying *GATE DA 2027 Prediction & Pattern Report* for the full reasoning.

General Aptitude (Q.1–Q.10)

Q.1 – Q.5 carry one mark each.

Q.1 [MCQ • 1 Mark]

Select the option that best completes the analogy.

Meticulous : Careless :: Voluble : _____

(A) Verbose (B) Taciturn (C) Fluent (D) Eloquent

Q.2 [MCQ • 1 Mark]

Select the grammatically correct completion.

Neither the manager nor the employees _____ satisfied with the revised policy.

(A) was (B) is (C) has been (D) were

Q.3 [NAT • 1 Mark]

The value of 15% of 40% of 500 is _____.

Q.4 [MCQ • 1 Mark]

Find the next term of the sequence 3, 8, 15, 24, 35, _____

(A) 44 (B) 46 (C) 50 (D) 48

Q.5 [MCQ • 1 Mark]

A sum of money doubles itself in 8 years under simple interest. The annual rate of interest is

(A) 8% (B) 12.5% (C) 25% (D) 10%

Q.6 – Q.10 carry two marks each.

Q.6 [MCQ • 2 Marks]

Two inlet pipes can fill a tank in 20 minutes and 30 minutes respectively, while an outlet pipe alone can empty the full tank in 60 minutes. If all three are opened together on an empty tank, the tank is filled in

(A) 10 minutes (B) 15 minutes (C) 18 minutes (D) 12 minutes

Q.7 [NAT • 2 Marks]

The mean mark of 40 students in a class is 60. Five students whose mean mark is 40 leave the class. The mean mark of the remaining students is _____ (rounded off to two decimal places).

Q.8 [MSQ • 2 Marks]

A wooden cube is painted on all six faces and then cut into 27 identical smaller cubes. Which of the following statement(s) is/are correct?

(A) Exactly 12 small cubes have exactly two painted faces

(B) Exactly 8 small cubes have three painted faces

(C) Exactly 4 small cubes have no painted face

(D) Exactly 6 small cubes have exactly one painted face

Q.9 [MCQ • 2 Marks]

The number of distinct arrangements of the letters of the word **SUCCESS** is

(A) 5040 (B) 840 (C) 210 (D) 420

Q.10 [MSQ • 2 Marks]

Consider the statements: *"If it rains, the match is cancelled."* and *"The match was not cancelled."* Which of the following conclusion(s) necessarily follow(s)?

(A) It rained

(B) If the match is cancelled, then it rained

(C) It did not rain

(D) "If the match is not cancelled then it did not rain" is a valid inference

Data Science & Artificial Intelligence (Q.11–Q.65)

Q.11 – Q.35 carry one mark each.

Q.11 [MSQ • 1 Mark]

Which of the following statement(s) about the Pearson chi-squared test is/are correct?

(A) For k categories with m parameters estimated from the data, the degrees of freedom are $k - 1 - m$

(B) The test is two-tailed

(C) The test statistic is $\sum (O_i - E_i)^2 / E_i$

(D) Expected cell frequencies should normally be at least 5

Q.12 [MCQ • 1 Mark]

A real square matrix Q is orthogonal if and only if

(A) $Q^{\mathsf{T}}Q = I$ (B) $\det(Q) = 1$ (C) $Q^{\mathsf{T}} = Q$ (D) $Q^2 = Q$

Q.13 [MCQ • 1 Mark]

In the ACID properties of a transaction, the letter **I** stands for

(A) Integrity (B) Indexing (C) Independence (D) Isolation

Q.14 [MSQ • 1 Mark]

Which of the following is/are standard OLAP operations on a data cube?

(A) Slice (B) Roll-up (C) Commit (D) Drill-down

Q.15 [MCQ • 1 Mark]

A schedule of transactions is conflict-serializable if and only if

(A) it contains no write operations

(B) its precedence (serialization) graph is acyclic

(C) every transaction in it uses two-phase locking

(D) it is a serial schedule

Q.16 [MCQ • 1 Mark]

The output of the logistic (sigmoid) activation function lies in

(A) $(0, 1)$ (B) $[0, \infty)$ (C) $[-1, 1]$ (D) $(-\infty, \infty)$

Q.17 [MCQ • 1 Mark]

A complete binary tree contains 15 nodes. Its height, measured as the number of edges on the longest root-to-leaf path, is

(A) 5 (B) 3 (C) 4 (D) 15

Q.18 [NAT • 1 Mark]

The Shannon entropy of a fair coin toss, measured in bits, is _____.

Q.19 [MCQ • 1 Mark]

If A is an $n \times n$ real matrix with $\det(A) = 0$, then

(A) the columns of A are linearly independent

(B) A is orthogonal

(C) $Ax = 0$ has a non-trivial solution

(D) A has n distinct eigenvalues

Q.20 [MCQ • 1 Mark]

For a dynamic array that doubles its capacity whenever it becomes full, the **amortized** cost of an append operation is

(A) $O(n \log n)$ (B) $O(n)$ (C) $O(1)$ (D) $O(\log n)$

Q.21 [MSQ • 1 Mark]

Which of the following supports lookup of an element in $O(1)$ **average** time?

(A) A balanced binary search tree

(B) An array, when the index is known

(C) A hash table with a good hash function

(D) A singly linked list

Q.22 [MCQ • 1 Mark]

For $f(x, y) = x^2y + y^3$, the value of $\dfrac{\partial f}{\partial x}$ at the point $(1, 2)$ is

(A) 2 (B) 4 (C) 13 (D) 14

Q.23 [MSQ • 1 Mark]

Which of the following is/are effective at reducing overfitting?

(A) Early stopping (B) L_2 regularization

(C) Increasing the number of hidden layers (D) Collecting more training data

Q.24 [MCQ • 1 Mark]

A 4×4 matrix has rank 2. The dimension of its null space is

(A) 2 (B) 0 (C) 1 (D) 3

Q.25 [MCQ • 1 Mark]

If $X \sim \text{Geometric}(p)$ taking values in $\{1, 2, 3, \dots\}$, then $E[X]$ equals

(A) $\dfrac{1}{p^2}$ (B) $\dfrac{1-p}{p}$ (C) $\dfrac{1}{p}$ (D) p

Q.26 [MCQ • 1 Mark]

If X and Y are independent random variables, then $\text{Var}(X - Y)$ equals

(A) $\text{Var}(X) + \text{Var}(Y)$ (B) $|\text{Var}(X) - \text{Var}(Y)|$

(C) $\text{Var}(X) - \text{Var}(Y)$ (D) $\text{Var}(X)\,\text{Var}(Y)$

Q.27 [NAT • 1 Mark]

For $R(A, B, C, D, E)$ with functional dependencies $A \to BC$, $CD \to E$ and $B \to D$, the number of attributes in the closure A^+ is _____.

Q.28 [NAT • 1 Mark]

The rank of the 3×3 matrix all of whose entries are equal to 1 is _____.

Q.29 [NAT • 1 Mark]

If $Z \sim N(0, 1)$, then $P(-1 < Z < 2) =$ _____ (rounded off to three decimal places).

Q.30 [NAT • 1 Mark]

The number of distinct truth assignments (models) over three propositional variables is _____.

Q.31 [MCQ • 1 Mark]

A biased coin with $P(\text{head}) = 0.4$ is tossed 10 times. The expected number of heads is

(A) 6 (B) 5 (C) 4 (D) 2.4

Q.32 [MCQ • 1 Mark]

For a search tree with branching factor b explored to depth d, the space complexity of depth-first

search is

(A) $O(b^{d/2})$ (B) $O(d)$ (C) $O(bd)$ (D) $O(b^d)$

Q.33 [MCQ • 1 Mark]

$\lim_{x \to 0} \dfrac{\sin 3x}{x}$ equals

(A) 0 (B) 3 (C) $\dfrac{1}{3}$ (D) 1

Q.34 [NAT • 1 Mark]

Consider the Python statement below.

```
print(len(1, 2, 2, 3, 3, 3))
```

The value printed is _____.

Q.35 [MCQ • 1 Mark]

The loss function normally minimised when training a binary classifier that outputs probabilities is

(A) hinge loss on the raw probabilities

(B) Kullback–Leibler divergence to a uniform prior

(C) binary cross-entropy

(D) mean squared error

Q.36 – Q.65 carry two marks each.

Q.36 [MSQ • 2 Marks]

Let A and B be $n \times n$ real matrices. Which of the following statement(s) is/are **always** correct?

(A) $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$

(B) Similar matrices have the same eigenvalues

(C) $\det(AB) = \det(A)\det(B)$

(D) The eigenvalues of $A + B$ are the sums of the eigenvalues of A and B

Q.37 [MSQ • 2 Marks]

Let f and g be convex functions on $\mathbb{R}^n$. Which of the following statement(s) is/are **always** correct?

(A) $f + g$ is convex

(B) Every local minimum of f is a global minimum

(C) fg is convex

(D) $\max\{f, g\}$ is convex

Q.38 [NAT • 2 Marks]

Let X be exponential with rate 1. The value of $E[X \mid X > 2]$ is _____.

Q.39 [NAT • 2 Marks]

A table `T(id, dept, sal)` contains the rows $(1, X, 100)$, $(2, X, 200)$, $(3, Y, 300)$, $(4, Y, 300)$, $(5, Z, 400)$. The number of rows returned by

```
SELECT dept FROM T
GROUP BY dept
HAVING COUNT(*) > 1;
```

is _____.

Q.40 [MCQ • 2 Marks]

The relation $R(A, B, C)$ has functional dependencies $AB \to C$ and $C \to A$. The highest normal form satisfied by R is

(A) 1NF only (B) 3NF (C) 2NF (D) BCNF

Q.41 [MCQ • 2 Marks]

The minimum value of $f(x, y) = x^2 + y^2 - 2x - 4y + 5$ over $\mathbb{R}^2$ is

(A) 0 (B) 1 (C) 5 (D) -1

Q.42 [MCQ • 2 Marks]

In a B$^+$ tree, a search key is 4 bytes and a block pointer is 8 bytes. If a node must fit in a 512-byte block, the maximum order n (number of block pointers per node) is

(A) 32 (B) 42 (C) 64 (D) 43

Q.43 [NAT • 2 Marks]

A two-ply game tree has a MAX node at the root with two MIN children. The leaf utilities under the two MIN nodes are $(5, 6, 7)$ and $(4, 9, 2)$ respectively. The minimax value at the root is _____.

Q.44 [MCQ • 2 Marks]

A directed acyclic graph has vertices $\{A, B, C, D\}$ and edges $A \to B$, $A \to C$, $B \to D$, $C \to D$. The number of distinct topological orderings is

(A) 2 (B) 6 (C) 4 (D) 1

Q.45 [NAT • 2 Marks]

The minimum value of $x^2 + y^2$ subject to the constraint $x + y = 10$ is _____.

Q.46 [MSQ • 2 Marks]

Which of the following statement(s) about a binary min-heap of n elements is/are correct?

(A) Extract-min takes $O(\log n)$ time

(B) An in-order traversal of the heap yields the elements in sorted order

(C) It can be built from an unsorted array in $O(n)$ time

(D) Heap sort can be performed in place

Q.47 [NAT • 2 Marks]

A logistic regression model has weight vector $w = (1, -1)$ and bias $b = 0$. For the input $x = (2, 1)$, the predicted probability of the positive class is _____ (rounded off to three decimal places).

Q.48 [MSQ • 2 Marks]

Which of the following statement(s) about principal component analysis is/are correct?

(A) The first principal component maximises the explained variance

(B) PCA uses the class labels to choose the components

(C) The principal components are mutually orthogonal

(D) The data should be mean-centred before applying PCA

Q.49 [NAT • 2 Marks]

A decision-tree node contains 6 positive and 4 negative examples. Its Gini impurity is _____ (rounded off to two decimal places).

Q.50 [MCQ • 2 Marks]

A box contains 10 items of which 3 are defective. Two items are drawn at random without replacement. The expected number of defective items drawn is

(A) 0.2 (B) 0.6 (C) 0.9 (D) 0.3

Q.51 [NAT • 2 Marks]

Let $T(n) = T(n/2) + 1$ with $T(1) = 0$, defined for n a power of 2. The value of $T(64)$ is _____.

Q.52 [MSQ • 2 Marks]

Which of the following statement(s) about the sampling distribution of the mean is/are correct?

(A) Increasing n narrows the confidence interval for μ

(B) $\mathrm{Var}(\bar{X}) = \sigma^2/n$ for i.i.d. observations

(C) The Central Limit Theorem requires $n > 1000$

(D) A 99% confidence interval is wider than a 95% confidence interval computed from the same sample

Q.53 [MSQ • 2 Marks]

Which of the following statement(s) about database indexes is/are correct?

(A) A clustered index determines the physical ordering of the rows

(B) A hash index answers range queries efficiently

(C) A table can have several clustered indexes

(D) A B^+ tree index answers range queries efficiently

Q.54 [MSQ • 2 Marks]

For $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, which of the following statement(s) is/are correct?

(A) The singular values of A are 3 and 1

(B) A is positive definite

(C) The eigenvalues of A are 3 and -1

(D) $\det(A) = -3$

Q.55 [MCQ • 2 Marks]

k-means with $k = 2$ is run on the one-dimensional points $\{1, 2, 3, 10\}$ with initial centroids 1 and 10. At convergence, the total within-cluster sum of squared distances is

(A) 4 (B) 0 (C) 1 (D) 2

Q.56 [MCQ • 2 Marks]

A node with 10 examples, 5 positive and 5 negative, is split into two children of 5 examples each, containing $(4^+, 1^-)$ and $(1^+, 4^-)$ respectively. The information gain of the split, in bits, is closest to

(A) 0.278 (B) 0.500 (C) 0.722 (D) 0.181

Q.57 [MCQ • 2 Marks]

Quick sort is run on an array that is already sorted in increasing order, always choosing the *first* element as the pivot. The running time is

(A) $\Theta(n^2)$ (B) $\Theta(n \log n)$ (C) $\Theta(\log n)$ (D) $\Theta(n)$

Q.58 [NAT • 2 Marks]

The system $x + y + z = 1$, $2x + 3y + kz = 3$, $x + ky + 3z = 2$ fails to have a unique solution for exactly two values of k. The larger of these two values is _____.

Q.59 [NAT • 2 Marks]

In a sample of 100 items, 55 possess a certain attribute. For testing $H_0 : p = 0.5$ against $H_1 : p \neq 0.5$, the value of the test statistic (using the null standard error) is _____ (rounded off to two decimal places).

Q.60 [MCQ • 2 Marks]

Let X_1, X_2, X_3 be i.i.d. Uniform$(0, 1)$ and $M = \max(X_1, X_2, X_3)$. Then $P(M < 0.5)$ equals

(A) 0.125 (B) 0.375 (C) 0.500 (D) 0.875

Q.61 [MCQ • 2 Marks]

Consider the Python code below.

```
x = 0
for i in range(1, 6):
for j in range(i, 6):
x += 1
print(x)
```

The value printed is

(A) 10 (B) 15 (C) 25 (D) 20

Q.62 [MSQ • 2 Marks]

Which of the following statement(s) about heuristics for A^* is/are correct?

(A) The heuristic $h(n) = 0$ is admissible

(B) A^* with an admissible heuristic is optimal in tree search

(C) Consistency implies admissibility

(D) Admissibility implies consistency

Q.63 [NAT • 2 Marks]

The number of structurally distinct binary search trees that can be built from 4 distinct keys is _____.

Q.64 [MCQ • 2 Marks]

The A^* algorithm with the heuristic $h(n) = 0$ for every node n behaves exactly like

(A) depth-first search (B) greedy best-first search (C) hill climbing (D) uniform-cost search

Q.65 [MCQ • 2 Marks]

Let $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. The set of eigenvalues of A^2 is

(A) $\{3, 1\}$ (B) $\{6, 2\}$ (C) $\{9, 1\}$ (D) $\{4, 1\}$

Answer Key

Answers only — no worked solutions. For NAT questions any value within the usual GATE tolerance of the figure shown is accepted. The **Subject** column tells you which topic to revise when you get one wrong. A key with several letters (e.g. **ABD**) is an MSQ.

Q	Ans	Subject	Q	Ans	Subject	Q	Ans	Subject
1	B	GA	**2**	D	GA	**3**	30	GA
4	D	GA	**5**	B	GA	**6**	B	GA
7	62.86	GA	**8**	ABD	GA	**9**	D	GA
10	CD	GA	**11**	ACD	PS	**12**	A	LA
13	D	DB	**14**	ABD	DB	**15**	B	DB
16	A	ML	**17**	B	PDSA	**18**	1	ML
19	C	LA	**20**	C	PDSA	**21**	BC	PDSA
22	B	CO	**23**	ABD	ML	**24**	A	LA
25	C	PS	**26**	A	PS	**27**	5	DB
28	1	LA	**29**	0.819	PS	**30**	8	AI
31	C	PS	**32**	C	AI	**33**	B	CO
34	3	PDSA	**35**	C	ML	**36**	ABC	LA
37	ABD	CO	**38**	3	PS	**39**	2	DB
40	B	DB	**41**	A	CO	**42**	D	DB
43	5	AI	**44**	A	PDSA	**45**	50	CO
46	ACD	PDSA	**47**	0.731	ML	**48**	ACD	ML
49	0.48	ML	**50**	B	PS	**51**	6	PDSA
52	ABD	PS	**53**	AD	DB	**54**	ACD	LA
55	D	ML	**56**	A	ML	**57**	A	PDSA
58	3	LA	**59**	1.00	PS	**60**	A	PS
61	B	PDSA	**62**	ABC	AI	**63**	14	PDSA
64	D	AI	**65**	C	LA			

Score sheet

Fill this in after checking the key. Apply GATE negative marking: $-1/3$ for a wrong 1-mark MCQ and $-2/3$ for a wrong 2-mark MCQ; MSQ and NAT carry no penalty.

Subject	Questions	Marks available	Marks scored	Time taken
General Aptitude	10	15		
Probability & Statistics	10	15		
Linear Algebra	8	12		
Calculus & Optimization	5	8		
Programming, Data Structures & Algorithms	10	16		
Database Management & Warehousing	8	12		
Machine Learning	9	14		
Artificial Intelligence	5	8		
Total	**65**	**100**		